

The Two-Time Moving-Defect Problem for Dispersing Billiards

A go/no-go audit and a centered two-sided bridge

Working note

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Abstract

We isolate the estimate needed to differentiate an infinite correlation susceptibility under a genuine motion of the scatterers. A tempting bridge from the Demers–Liverani projective cone to forward standard families fails: the moving-face coefficient uses a one-sided trace on a transversal terminal arc, whereas the cone controls stable-leaf integrals. We give a local counterexample to any inference of such a trace bound from the published cone seminorms. We then formulate the correct centered kernel problem and prove a two-sided nuclear/recovery criterion which yields the desired product estimate. The remaining billiard-specific task is reduced to a blockwise factorization of the complete physical defect, including a uniform difference-quotient version. No unconditional CM2 theorem is claimed here.

1 The exact target

Let P_s be the Markov-normalized transfer operator on a fixed collision fiber, $P_s \mathbf{1} = \mathbf{1}$, let

$$\Pi h = \mu_0(h) \mathbf{1}, \quad Q = P_0 - \Pi,$$

and let K_V denote the complete first moving-table defect after all regular terms, same-label physical faces, and grazing blocks have been assembled. The two-time estimate needed for a prescribed centered source g is

$$|\langle Q^n K_V Q^m g, f \rangle| \leq C \tau^m \vartheta^n \|g\|_X \|f\|_{C^1}, \quad m, n \geq 0, \quad (1)$$

for some $\tau, \vartheta < 1$. The same estimate, uniformly for $s^{-1}(P_s - P_0)$, is required before a parameter limit can be exchanged with the two time sums.

The operator must be centered as a whole:

$$K_V = \Pi^\perp \partial_s P_s|_{s=0} \Pi^\perp. \quad (2)$$

Centering separate faces is neither invariant nor sufficient. Without (2), rank-one terms generally give an additive bound $O(\tau^m + \vartheta^n)$, whose double sum diverges.

2 Why the direct projective-cone bridge fails

The projective cone for dispersing billiards controls integrals of a density against positive dynamically Hölder tests on stable curves, short-leaf mass, and Hölder comparison of

such integrals between nearby stable curves. A moving inverse-branch boundary instead produces, before physical assembly, a coefficient of the form

$$h \longmapsto w(h|_{I^+} - h|_{I^-}) \delta_\Gamma, \quad (3)$$

where $I^\pm$ are one-sided terminal preimage arcs. In the radial collision atlas these arcs land on the grazing seam and are transversal to the stable test foliation. Thus (3) asks for a transverse trace, not a stable-leaf average.

The following elementary local model rules out the missing implication.

Proposition 2.1 (Stable-leaf control does not dominate transverse trace). *Let $R = (-2, 2)^2$, let the local stable leaves be the horizontal segments $W_y = (-2, 2) \times \{y\}$, and let the transversal face be $I = \{0\} \times (-1, 1)$. Fix $0 < q < 1$ and $0 < \gamma < 1$. There are smooth functions u_ε supported in an ε -tube about I such that*

$$u_\varepsilon|_I = b \neq 0,$$

while, uniformly over bounded Hölder test functions ψ on stable subsegments $W \subset W_y$,

$$\left| \int_W u_\varepsilon \psi \, dm_W \right| \leq C \min\{|W|, \varepsilon\} \|\psi\|_\infty, \quad (4)$$

$$\sup_{0 < |W| \leq 1} |W|^{-q} \left| \int_W u_\varepsilon \psi \, dm_W \right| \leq C \varepsilon^{1-q} \|\psi\|_\infty, \quad (5)$$

and the Hölder comparison of these integrals between nearby parallel stable leaves is $O(\varepsilon)$. Consequently no norm or order estimate using only the stable-leaf quantities in (4)–(5) and their transverse Hölder comparisons can bound $\|u|_I\|_{C^0}$.

Proof. Choose $\phi \in C_c^\infty((-1, 1))$ with $\phi(0) = 1$ and $b \in C_c^\infty((-1, 1))$, $b \neq 0$, and set

$$u_\varepsilon(x, y) = \phi(x/\varepsilon)b(y).$$

The trace assertion is immediate. A horizontal subsegment meets the support in length at most $C \min\{|W|, \varepsilon\}$, proving (4). If $|W| \leq \varepsilon$, multiplication by $|W|^{-q}$ gives at most $C \varepsilon^{1-q}$; if $|W| \geq \varepsilon$, it gives at most $C \varepsilon |W|^{-q} \leq C \varepsilon^{1-q}$, proving (5). Replacing y by y' changes the integral by at most $C \varepsilon |y - y'| \|b\|_{C^1} \|\psi\|_{C^0}$, with the usual additional Hölder test-comparison term. The trace is independent of ε , so the asserted domination is impossible. $\square$

Corollary 2.2 (Failure of the proposed positive cone operator). *The local face map (3) is not known to be positive for the Demers–Liverani cone order, and its boundedness cannot be inferred from projective contraction. In particular, the implication*

$$\Theta(P_0^m f_+, P_0^m f_-) \leq C \chi^m \implies \|K_V P_0^m (f_+ - f_-)\|_Y \leq C' \chi^m$$

is invalid without an additional transverse-trace theorem.

Proof. The cone order is tested through stable-leaf integrals. The face coefficient contains precisely the trace excluded by Proposition 2.1. Projective contraction therefore supplies no order comparison for the two output face measures. $\square$

This is a go/no-go conclusion, not merely a missing estimate: the direct “past cone $\rightarrow$ face $\rightarrow$ future family” proof must be abandoned unless physical assembly first converts the terminal trace into an integral functional controlled by the cone. The existing adjoint dyadic bound does not perform that conversion.

3 The correct two-sided bridge

The complete defect is a kernel located at the event time. It must be controlled from the past and the future simultaneously. The following criterion is abstract but exact.

Definition 3.1 (Centered nuclear recovery bridge). *Let X_- be a Banach space of centered past observables and let Y_+ be a Banach space of centered signed measured unstable families. A centered operator $K : X_- \rightarrow Y_+$ has a nuclear recovery presentation if*

$$K = \sum_{j \geq 1} \ell_j \otimes \eta_j, \quad \sum_{j \geq 1} \|\ell_j\|_{X_-^*} \|\eta_j\|_{Y_+} < \infty, \quad (6)$$

where $\ell_j(\mathbf{1}) = 0$ and $\eta_j(\mathbf{1}) = 0$ after the complete regular and physical-face assembly.

Theorem 3.2 (Two-sided bridge implies CM2). *Assume*

$$\|Q^m g\|_{X_-} \leq C_- \tau^m \|g\|_{X_-}, \quad (7)$$

$$|\eta(\mathcal{U}_0^n f)| \leq C_+ \vartheta^n \|\eta\|_{Y_+} \|f\|_{C^1}, \quad (8)$$

for centered g and centered η . If K has the presentation (6), then

$$|(KQ^m g)(\mathcal{U}_0^n f)| \leq C_- C_+ \mathfrak{N}(K) \tau^m \vartheta^n \|g\|_{X_-} \|f\|_{C^1},$$

where $\mathfrak{N}(K) = \sum_j \|\ell_j\|_{X_-^*} \|\eta_j\|_{Y_+}$.

Proof. Using (6), (7), and (8),

$$\begin{aligned} |(KQ^m g)(\mathcal{U}_0^n f)| &\leq \sum_j |\ell_j(Q^m g)| |\eta_j(\mathcal{U}_0^n f)| \\ &\leq C_- C_+ \tau^m \vartheta^n \|g\|_{X_-} \|f\|_{C^1} \sum_j \|\ell_j\|_{X_-^*} \|\eta_j\|_{Y_+}. \end{aligned}$$

□

Remark 3.3. *The theorem makes clear why a usual functional correlation estimate with error $O(\tau^m + \vartheta^n)$ is insufficient. CM2 requires a centered operator ideal estimate at the event time, not just mixing on either side. Wormell's specialized two-time conditional-mixing conjecture is one way to encode precisely this product structure.*

4 Recovery-time weighted future space

For a measured unstable component (W, ρ) let $R(W)$ be the first time at which its canonical forward subdivision is a proper standard family. For $\omega > 0$, define schematically

$$\|\eta\|_{Y_+} := \inf \sum_j |a_j| e^{\omega R(W_j)} (1 + \text{Reg}(\rho_j)), \quad \eta = \sum_j a_j (W_j, \rho_j). \quad (9)$$

The infimum also allows the positive and negative Jordan parts to be split by recovery rank. Standard-family coupling and recovery give (8) once the constants in (9) are chosen within the recovery tail.

For a radial grazing block b , the physical flux factorization gives a mass bound $M_b \leq C2^{-b}$. If its geometric deterioration is polynomial in 2^b , the recovery lemmas give

$$R_b \leq C_0 + C_1 b.$$

Hence

$$\sum_{b \geq 0} M_b e^{\omega R_b} < \infty \quad \text{whenever} \quad \omega C_1 < \log 2. \quad (10)$$

Thus initial finite Z is unnecessary. What remains is not future recovery, but the past functional ℓ_j in (6).

5 Billiard-specific block problem

For each genuine physical block b , the complete difference quotient must be written, after artificial-face cancellation, as

$$K_{V,b} = \sum_{k \geq 1} \ell_{b,k} \otimes \eta_{b,k}, \quad (11)$$

with

$$\sum_{b,k} \|\ell_{b,k}\|_{X_-^*} \|\eta_{b,k}\|_{Y_+} < \infty. \quad (12)$$

The desired estimate follows from Theorem 3.2. There are only three honest possibilities for proving (12):

- (i) a reverse-time conditional-mixing theorem for the terminal slice, furnishing X_-^* directly;
- (ii) a two-sided Young-tower lift of the complete centered defect, with an event kernel that is nuclear between the past and future tower norms;
- (iii) a smoothing/nuclear decomposition of the terminal restriction, with a quantitative bunching inequality strong enough to sum the smoothing error over the homogeneity blocks.

The projective-cone contraction alone supplies none of these three inputs.

5.1 A two-sided tower formulation

The natural extension gives a sharper version of alternative (ii). Let Δ be a two-sided Young tower for the fixed billiard, with past and future cylinder filtrations $\mathcal{F}_i^-$ and $\mathcal{F}_j^+$. Denote the associated martingale differences by

$$D_i^- = \mathbb{E}(\cdot \mid \mathcal{F}_i^-) - \mathbb{E}(\cdot \mid \mathcal{F}_{i-1}^-), \quad D_j^+ = \mathbb{E}(\cdot \mid \mathcal{F}_j^+) - \mathbb{E}(\cdot \mid \mathcal{F}_{j-1}^+).$$

A lifted event kernel $\widehat{K}$ has summable mixed cylinder variation if

$$\sum_{i,j \geq 1} \|D_i^- D_j^+ \widehat{K}\|_{\mathcal{B}_-^* \widehat{\otimes}_\pi \mathcal{B}_+} < \infty. \quad (13)$$

The indices start at one: complete double centering removes the two axis terms $D_i^- D_0^+ \widehat{K}$ and $D_0^- D_j^+ \widehat{K}$.

Proposition 5.1 (Mixed-cylinder criterion). *Suppose the two-sided tower has exponential tails and the past and future transfer actions contract the centered martingale blocks at rates τ^m and ϑ^n . If the complete lifted moving defect satisfies (13), then it has a nuclear recovery presentation (6) and hence satisfies CM2.*

Proof. Expand

$$\widehat{K} = \sum_{i,j \geq 1} D_i^- D_j^+ \widehat{K}$$

in the projective tensor product. Each summand is a norm-convergent sum of rank-one past/future tensors by the definition of $\widehat{\otimes}_\pi$. Summability in (13) therefore gives (6). Shifting m steps into the past and n steps into the future contracts the two centered tensor factors separately. Apply Theorem 3.2. $\square$

This criterion identifies the exact geometric issue. A smooth two-dimensional density on a tower rectangle has summable mixed cylinder variation. A face current is instead supported on a one-dimensional graph in the local stable–unstable product chart. Its graph delta is not automatically in the projective tensor product. Proving (13) is therefore a genuine conditional-mixing theorem, not routine tower bookkeeping. The finite-strip central face supplies uniform transversality and one-sided recovery, but these facts alone do not prove the mixed tensor estimate.

5.2 A quantitative smoothing test

There is one falsifiable way to obtain (13). Replace a face delta in block b by a normalized collar of width ϵ . Suppose the regularized bridge and the collar error satisfy

$$\|K_{b,\epsilon}\|_{X_- \rightarrow Y_+} \leq CM_b \epsilon^{-p}, \quad (14)$$

$$|((K_{V,b} - K_{b,\epsilon})Q^m g)(\mathcal{U}_0^n f)| \leq CM_b \epsilon^\alpha \Lambda_-^m \Lambda_+^n \|g\| \|f\|_{C^1}. \quad (15)$$

If the regularized dynamics gives rates $\tau_0^m \vartheta_0^n$, choosing $\epsilon = e^{-a(m+n)}$ yields CM2 provided one can choose $a > 0$ so that

$$\tau_0 e^{ap} < 1, \quad \vartheta_0 e^{ap} < 1, \quad \Lambda_- e^{-a\alpha} < 1, \quad qquad \Lambda_+ e^{-a\alpha} < 1. \quad (16)$$

Equivalently, a genuine spectral/bunching margin is required. For billiards, unbounded grazing derivatives make (15) the difficult part. The current flux estimate $M_b \lesssim 2^{-b}$ and recovery bound $R_b \lesssim b$ control the sum over b , but do not supply the normal Hölder estimate in (15). Thus collar smoothing is a testable fallback, not a completed proof.

Proposition 5.2 (Uniform difference-quotient closure). *Suppose (11)–(12) hold uniformly with $K_{V,b}$ replaced by the assembled block of $s^{-1}(P_s - P_0)$, and each block converges weakly to $K_{V,b}$. Then the parameter difference quotient of the prescribed susceptibility is dominated by a summable $C\tau^m \vartheta^n$ majorant and converges term by term to the double series generated by K_V .*

Proof. Apply Theorem 3.2 uniformly in s and sum over b, m, n . Equation (12) and the two geometric series give an integrable majorant. Weak convergence on every block identifies the pointwise limit; dominated convergence completes the argument. $\square$

6 Next falsifiable test

The next calculation should use one central physical face and a finite strip cutoff. It must decide whether its complete coarea formula admits a decomposition (11) with a past functional that is an *integral* on a reverse-admissible curve. If the coefficient remains a naked one-sided value on the terminal preimage arc, Proposition 2.1 rules out the projective-cone

route and the proof must proceed through a reverse conditional slice or a two-sided tower. Only after this local test passes should the dyadic estimate (10) be added.

The current status is therefore precise: future recovery is technically available, complete centering is algebraic, and the prescribed source is already independent and non-coboundary. The unresolved theorem is the past-side terminal-slice bound in (12), equivalently the mixed cylinder estimate (13). Calling this bound a consequence of the existing projective cone would be incorrect.

References

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